A Trace Bound for Negative Theta-Form Spectrum

Jongmin Choi
Independent Researcher, Seoul, Korea
ORCID: 0009-0008-7448-514X
24ping@naver.com

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Abstract

For the actual centered theta–Gamma–prime form, and conditional on its stated self-adjoint form realization, we bound the negative spectral count by the trace of the negative part of a finite prime-head Fourier multiplier plus the complete tail norm. The bound uses the actual centered weighted Gram and admits additional exact finite radical constraints. It gives $N_-(\delta) \le \delta^{-1}\exp(O(\sqrt{\log(1/\delta)}))$ as $\delta \downarrow 0$, and the negative part of the centered operator belongs to every Schatten class $\mathfrak{S}_p$ for $p > 1$. The trace-count principle is classical; the content here is its application with the stated actual form, full tail, and constants. No assertion of a negative eigenvalue, global positivity, trace-class membership at $p = 1$, or a proof or disproof of the Riemann hypothesis is made.

Keywords: theta kernel; prime powers; negative spectrum; trace estimates; Schatten ideals.

1 The complete form and its finite head

For $x \ge 0$, set $\tau_n(x) = \pi n^2 e^{2x}$ and define the positive even theta kernel and its weight by

$$\kappa(x) = e^{x/2}\sum_{n\ge1}(4\tau_n^2 - 6\tau_n)e^{-\tau_n}, \qquad W(x) = \frac{\kappa(x)}{2\cosh(x/2)}.$$

Work in the complex even weighted space $\mathcal{H} = L^2_{\mathrm{even}}(dx/W)$, with inner product conjugate-linear in the first argument. Put $m(u) = \int \cosh(x/2)u(x)dx$, $\mathcal{H}_c = \ker m$, and $Cu = u - 2m(u)\kappa$, the $\mathcal{H}$-orthogonal projection onto $\mathcal{H}_c$. The exact theta normalization is $\|\kappa\|_{\mathcal{H}} = 1$ and $\int 2\cosh(x/2)\kappa(x)dx = 1$; hence

$$\int_{\mathbb{R}} W(x)dx \le \tfrac{1}{4}. \tag{1}$$

Use the unitary ordinary Fourier transform and write $\|\cdot\|_2$ for ordinary $L^2(dx)$. Define

$$\rho(s) = \frac{e^{-s/2}}{1-e^{-2s}}, \qquad t_\Gamma[u] = \int_0^\infty \rho(s)\|u(\cdot+s) - u\|_2^2 ds = \int_{\mathbb{R}} d_\Gamma(t)|\widehat{u}(t)|^2 dt,$$

$$d_\Gamma(t) = 2\int_0^\infty \rho(s)(1-\cos ts)ds \ge \tfrac{1}{2}\log(1+4t^2),$$

$$p[u] = 2\,\mathrm{Re}\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\int_{\mathbb{R}} u(x+\log n)\overline{u(x)}dx, \qquad q_0[u] = t_\Gamma[u] - c_\Gamma\|u\|_2^2 - p[u],$$

$$c_\Gamma = \log\pi - \psi(1/4) = \log\pi + \gamma_E + \pi/2 + 3\log 2 \in (0,6).$$

The lower bound on d_Γ follows from $1 - e^{-2s} \le 2s$; the special-value formula DLMF 5.4.19 gives the identity for c_Γ [5]. All prime powers and both orientations are included.

We use the exact centered form domain

$$X_c = \left\{ u \in L^2_{\mathrm{even}}(dx/W) : \ m(u) = 0, \quad \int_{\mathbb{R}} d_\Gamma(t) |\widehat{u}(t)|^2 dt < \infty \right\}.$$

As an input from the accompanying working manuscript [1], q_0 is a densely defined, closed, lower-bounded form on X_c in $\mathcal{H}_c$, the complete theta/prime correction is bounded, and L_c is its form-defined self-adjoint representative. That source also establishes the actual derivative radicals $r_j = \kappa^{(2j)} - 4^{-j}\kappa \in \ker L_c$, which are linearly independent. Their closed span $\mathcal{N}$ is contained in, but not asserted equal to, the full physical kernel. The form realization is an input; no identity with a separately defined minimal symmetric closure is asserted.

For integer $N \ge 1$, write

$$c_N(t) = c_\Gamma + 2 \sum_{2 \le n \le N} \frac{\Lambda(n)}{\sqrt{n}} \cos(t \log n), \qquad b_N(t) = (c_N(t) - d_\Gamma(t))_+. \tag{2}$$

Let $q_N = q_0 + p_{>N}$ be the head form. On X_c its exact ordinary Fourier expression is

$$q_N[u] = \int_{\mathbb{R}} (d_\Gamma(t) - c_N(t)) |\widehat{u}(t)|^2 dt. \tag{3}$$

The entire prime remainder is represented on $\mathcal{H}_c$ by a bounded self-adjoint operator of norm at most [2]

$$\tau_N := 80 \sum_{n > N} \Lambda(n) n^{3/2} e^{-\pi n} \le 160 (N+1)^2 e^{-\pi(N+1)}. \tag{4}$$

Thus no tail sign is assumed. The standard trace-count principle used below is classical; see, for example, [6]. The purpose here is its application to this particular full theta form and its actual centered Gram.

2 A positive trace-class head deficit

For $t \ge 0$ let $g_t(x) = W(x) \cos(tx)/\sqrt{2\pi}$. For even $u \in \mathcal{H}$, $\widehat{u}(t) = \langle g_t, u \rangle_{\mathcal{H}}$. Set

$$C_N = c_\Gamma + 2 \sum_{2 \le n \le N} \frac{\Lambda(n)}{\sqrt{n}}, \qquad T_N = \tfrac{1}{2} \sqrt{e^{2C_N} - 1}. \tag{5}$$

Then $0 \le b_N \le C_N$ and $b_N(t) = 0$ for $|t| \ge T_N$ by the Gamma lower bound. Define on $\mathcal{H}_c$ the positive operator

$$B_N = 2 \int_0^{T_N} b_N(t) \, (Cg_t)(Cg_t)^* dt. \tag{6}$$

Here vv^* maps u to $v\langle v, u \rangle$; the integral is a positive trace-norm integral because $\int b_N \|Cg_t\|^2_{\mathcal{H}} dt < \infty$. In particular B_N is trace class and, for all $u \in \mathcal{H}_c$,

$$\langle u, B_N u \rangle_{\mathcal{H}} = \int_{\mathbb{R}} b_N(t) |\widehat{u}(t)|^2 dt. \tag{7}$$

The actual centered trace is

$$\operatorname{tr} B_N = 2 \int_0^{T_N} b_N(t) \|Cg_t\|^2_{\mathcal{H}} dt = 2 \int_0^{T_N} b_N(t) \left[\frac{1}{2\pi} \int_{\mathbb{R}} W(x) \cos^2(tx) dx - |\widehat{\kappa}(t)|^2 \right] dt. \tag{8}$$

The square bracket is nonnegative because it is a true weighted projection norm. In particular, by (1),

$$\operatorname{tr} B_N \le \frac{T_N - \frac{1}{2}\arctan(2T_N)}{4\pi} < \frac{e^{C_N}}{8\pi}. \tag{9}$$

Indeed $b_N(t) \le [C_N - \frac{1}{2}\log(1+4t^2)]_+$, and the integral of this bound from zero to T_N is $T_N - \frac{1}{2}\arctan(2T_N)$. The centering term $|\widehat{\kappa}|^2$ is retained in (8); the last displayed estimate discards it only in the upper direction.

Additional *finite* exact radical constraints may improve this bound. For $j \ge 0$, let $R_j = (r_1, \dots, r_j)$, $H_j = R_j^* R_j$ (the true weighted Gram), and $Q_j = I - R_j H_j^{-1} R_j^*$ on $\mathcal{H}_c$; for $j = 0$ put $Q_0 = I$. Define $B_{N,j} = Q_j B_N Q_j$. Then

$$\operatorname{tr} B_{N,j} = 2\int_0^{T_N} b_N(t)\left(\|Cg_t\|_{\mathcal{H}}^2 - v_j(t)^* H_j^{-1} v_j(t)\right) dt, \qquad v_j(t) = R_j^* C g_t. \tag{10}$$

For $j = 0$ the Gram correction is zero. For $j \ge 1$ the inverse is well-defined because the listed radicals are independent. This exact formula uses the actual centered and radical Grams; it does not replace them by identity coefficient matrices or assume the known radical is the entire kernel.

3 The negative spectral count

For $\delta > 0$ define the count, including multiplicity and the boundary,

$$N_-(\delta) = \dim \operatorname{Ran} \mathbf{1}_{(-\infty,-\delta]}(L_c). \tag{11}$$

Initially this dimension may be infinite. The endpoint convention is essential: eigenvalues equal to $-\delta$ are included.

Theorem 3.1. *For every integer $N \ge 1$, every finite radical bank $j \ge 0$, and every $\delta > \tau_N$,*

$$N_-(\delta) \le \dim \operatorname{Ran} \mathbf{1}_{[\delta-\tau_N,\infty)}(B_{N,j}) \le \frac{\operatorname{tr} B_{N,j}}{\delta - \tau_N} \le \frac{\operatorname{tr} B_N}{\delta - \tau_N}. \tag{12}$$

Consequently the negative spectrum consists of isolated eigenvalues of finite multiplicity, with no negative accumulation point except possibly zero.

Proof. Since $d_\Gamma - c_N \ge -b_N$ pointwise, (3), (7), and the *complete* tail norm give the form inequality

$$q_0[u] \ge -\langle u, B_N u\rangle_{\mathcal{H}} - \tau_N \|u\|_{\mathcal{H}}^2 \qquad (u \in X_c). \tag{13}$$

Lower boundedness puts every vector in $\operatorname{Ran} \mathbf{1}_{(-\infty,-\delta]}(L_c)$ in $\operatorname{Dom} L_c$. Every such vector has $q_0[u] \le -\delta\|u\|_{\mathcal{H}}^2$ and is orthogonal to the entire zero eigenspace, hence belongs to $\operatorname{Ran} Q_j$. Thus on that whole spectral subspace

$$\langle u, B_{N,j} u\rangle_{\mathcal{H}} \ge (\delta - \tau_N)\|u\|_{\mathcal{H}}^2.$$

If the negative spectral subspace had larger dimension than the spectral subspace of $B_{N,j}$ on $[\delta-\tau_N, \infty)$, it would contain a nonzero vector orthogonal to that latter subspace. Its $B_{N,j}$ Rayleigh quotient would be strictly smaller than $\delta - \tau_N$, contradicting the displayed inequality. The spectral theorem and positivity bound the latter dimension by $\operatorname{tr} B_{N,j}/(\delta - \tau_N)$, proving (12), even before finite dimensionality of the negative subspace is known. Since $\tau_N \to 0$, the count is finite at every $\delta > 0$. The lower-bounded spectral theorem then gives the stated discrete-spectrum conclusion. □

This argument is the classical trace-count estimate; it neither evaluates B_N numerically nor proves a sign of L_c. The tail contributes with its full norm, not with a presumed favorable sign.

4 Explicit cutoff and Schatten consequence

An elementary Chebyshev/partial-summation bound gives [3]

$$\sum_{2\le n\le N}\frac{\Lambda(n)}{\sqrt{n}}\le 4(\log 2)\sqrt{N}+5,\qquad C_N\le C_N^*:=16+8(\log 2)\sqrt{N}.$$

For completeness, if $\psi_{\mathrm{C}}(x)=\sum_{n\le x}\Lambda(n)$, the binomial estimate $\psi_{\mathrm{C}}(n)\le\psi_{\mathrm{C}}(\lceil n/2\rceil)+n\log 2$ implies $\psi_{\mathrm{C}}(x)\le 2(\log 2)x+\log x+\log 2$, and partial summation gives the displayed bound. This ψ_{C} is distinct from the digamma ψ.

For $0<\delta\le 1$, choose

$$N_\delta=\left\lceil 3+\log\frac{2}{\delta}\right\rceil. \tag{14}$$

The upper bound in (4) gives $\tau_{N_\delta}<\delta/2$. Indeed for $L=\log(2/\delta)$, $4+L\le N_\delta+1\le 5+L$, and $\pi>3$ yields

$$\tau_{N_\delta}\le 160(5+L)^2e^{-12-3L}\le 4000e^{-12}e^{-L}<e^{-L}=\delta/2.$$

Here $(1+L/5)^2e^{-2L}\le 1$ and $e^{12}>4096>4000$. Combining (12) and (9) gives the explicit bound

$$N_-(\delta)<\frac{e^{C_{N_\delta}^*}}{4\pi\delta}. \tag{15}$$

It follows, with a constant depending only on the stated actual input bounds, that

$$N_-(\delta)\le\delta^{-1}\exp\bigl(O(\sqrt{\log(1/\delta)})\bigr)\qquad(\delta\downarrow 0). \tag{16}$$

The estimate is a one-sided upper bound and no optimality or numerical usefulness is claimed.

Corollary 4.1. *The negative part $(L_c)_-=(-L_c)_+$ is compact and belongs to $\mathfrak{S}_p$ for every $p>1$.*

Proof. By Theorem 3.1, the negative part has eigenvalues tending only to zero. Its eigenvalues $\mu_n>0$ satisfy $N_-(\delta)=\#\{n:\mu_n\ge\delta\}$. Since L_c is lower bounded,

$$\sum_n\mu_n^p=p\int_0^\infty\delta^{p-1}N_-(\delta)d\delta.$$

The estimate (16) makes the integral near zero finite for $p>1$: after $x=\log(1/\delta)$ it is bounded by a constant times $\int^\infty e^{-(p-1)x+O(\sqrt{x})}dx$. The remaining interval is finite by Theorem 3.1. The argument gives no conclusion for $p=1$. □

5 Scope and sources

The underlying working manuscript [1] supplies the precise theta normalization, form-defined centered representative, Fourier Gamma lower bound, bounded complete correction, and actual radicals (notably Lemmas a:lem:theta, a:lem:prime, a:lem:fourier and a:lem:radical). The supplied working notes [2, 3, 4] record the complete tail, elementary prime caps, and the related trace-count argument. The analytical argument here does not rely on the separate numerical statements in those records. The general trace-count principle predates this application [6]; no historical priority is claimed. The accompanying finite checks verify elementary inequalities and matrix algebra, not the full analytic realization or a numerical spectral certificate.

The bounds do not determine whether $N_-(\delta)$ is zero or positive for any δ, establish $\mathcal{N}=\ker L_c$, settle $\mathfrak{S}_1$ membership, or prove or disprove the Riemann hypothesis. No floating-point $N=2$ scalar bracket is used; no numerical trace evaluation is claimed.

References

[1] J. Choi, *Theta Prime Gamma Energy and its Spectral Structure*, version 2, November 2026, TeX and PDF snapshot supplied; immutable repository commit `91e8742de9b03fd363c1c6b2c25363dc861c4515`, `https://github.com/serabing-hash/theta-weighted-certificate/tree/91e8742de9b03fd363c1c6b2c25363dc861c4515/papers/2026-11-03-unified-theta-energy-v2`.

[2] *Every finite prime head has infinite signed index on the actual radical*, research note, 3 November 2026, supplied.

[3] *Explicit low-collar prime-tail compression and finite radical certificates*, research note, 3 November 2026, supplied.

[4] *Negative spectral counts from the finite-head scalar trace*, analytical research note, 3 November 2026, supplied. Its separate numerical claims are not used here.

[5] NIST Digital Library of Mathematical Functions, equations 5.4.19 and 5.9.16, `https://dlmf.nist.gov/5.4.E19`, `https://dlmf.nist.gov/5.9.E16`.

[6] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., Mathematical Surveys and Monographs 120, American Mathematical Society, 2005, `https://doi.org/10.1090/surv/120`.